

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T57 · Venn and Tree Diagrams

With Answers

Questions with full worked solutions

■ 40 pages

FAMILY MAP

Probability Diagrams Venn Tree Diagrams

F01

Construct and read probability trees

7 Q | 18 marks

F02

Represent and read events with Venn diagrams

8 Q | 35 marks

Working Unit 10

4MA1-1H Q14

ORIGINAL QUESTION**4 marks**

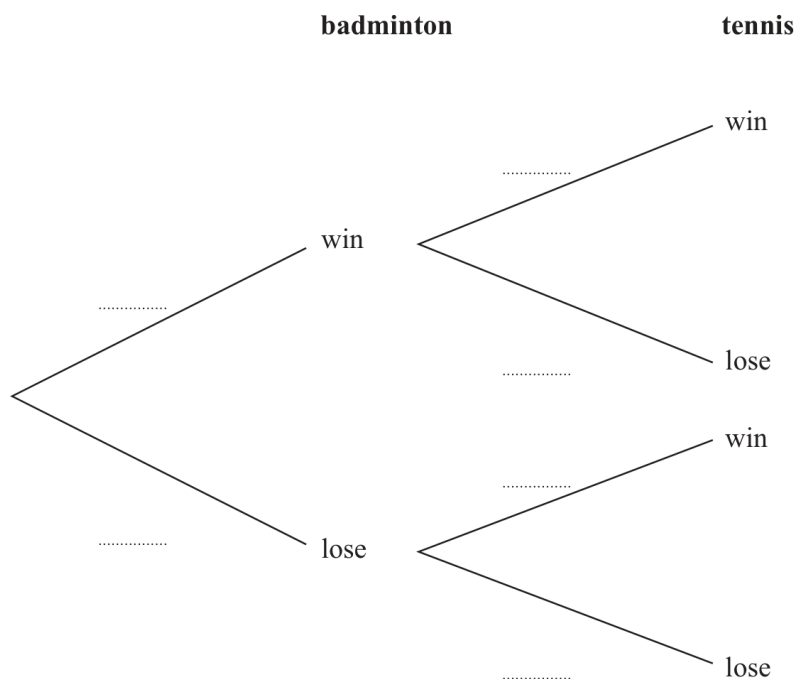
- 14** Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that Ricardo loses both games.

.....
(2)

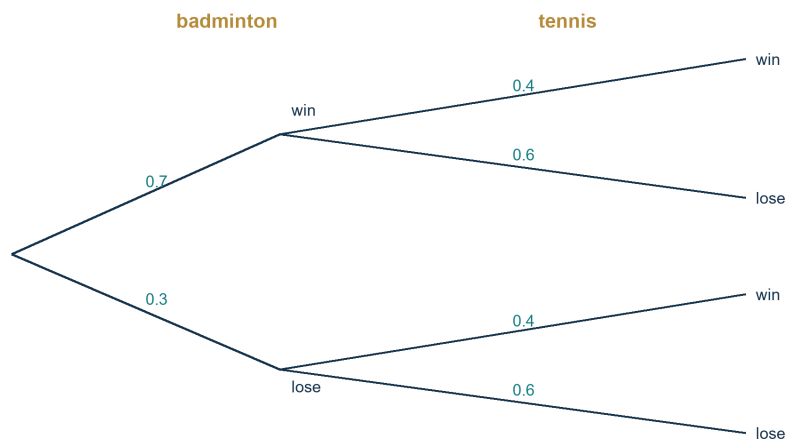
Worked Solution 10

Complete a probability tree

UNIT 1
4 marks**Complementary branches**

$$P(B_W) = 0.7, P(B_L) = 0.3; \quad P(T_W) = 0.4, P(T_L) = 0.6 \text{ after either first result.}$$
Required path

$$P(\text{lose both}) = 0.3 \times 0.6 = 0.18.$$



Solved reference diagram

Final answer

0.18

Dependency check Part (b) reads the two loss branches from the completed tree.

Question 24

4MA1-2H Q16

ORIGINAL QUESTION

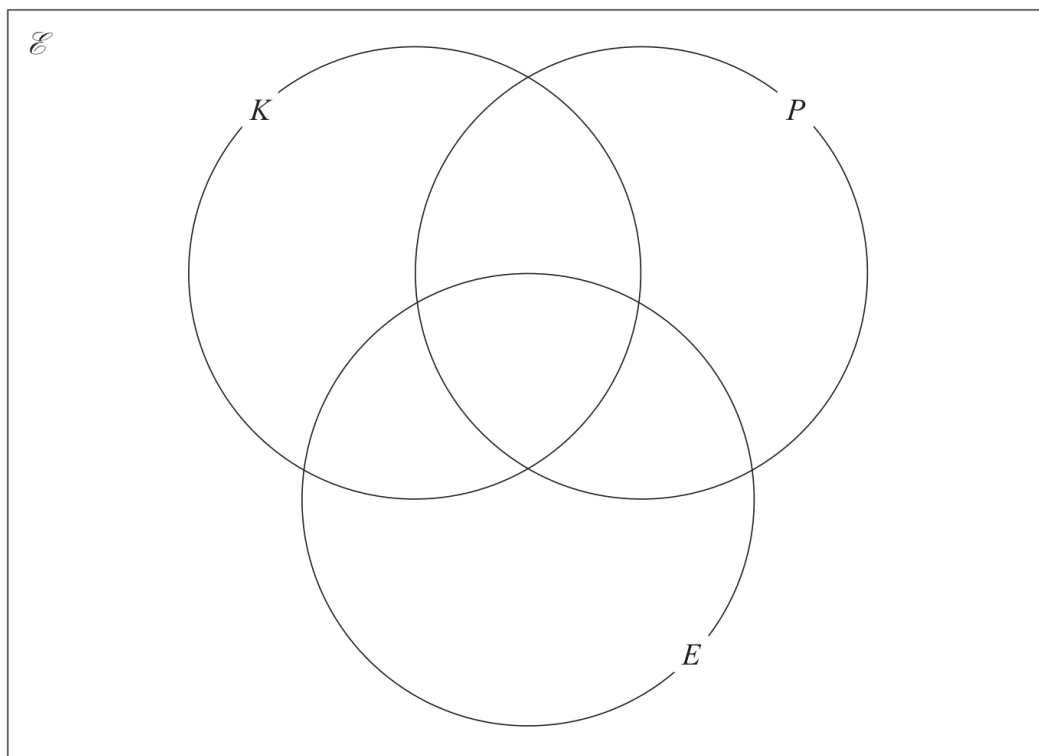
7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

Question 24 (continued)

4MA1-2H Q16

ORIGINAL QUESTION

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

(2)

(c) Find $n(P \cap K')$

(1)

(d) Find $n([P \cup E] \cap K)$

(1)

Worked Solution 24

Three-set Venn diagram

MODULAR UNIT 1

7 marks

Regions

$$KPE = 9, KP = 8, PE = 7, KE = 11, P = 4, E = 12, K = 7, \text{ outside} = 2.$$

Conditional probability

$$P(K | P) = 17/28.$$

Counts

$$n(P \cap K') = 4 + 7 = 11; \quad n((P \cup E) \cap K) = 8 + 11 + 9 = 28.$$

Final answer

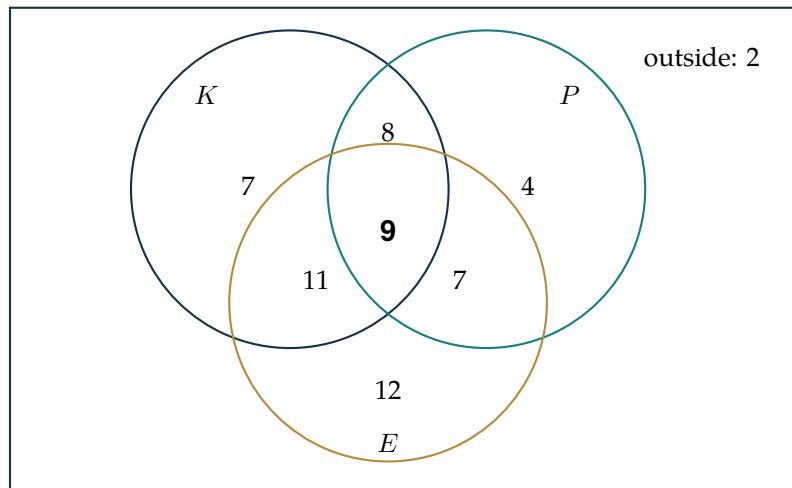
$$\frac{17}{28}; \quad 11; \quad 28$$

Completed diagram 24

Three-set Venn diagram

MODULAR UNIT 1

7 marks

Completed solution diagram: Completed three-set Venn diagram

Original question 10

4MA1-1H Q13 M01, M02

MODULAR UNIT 1

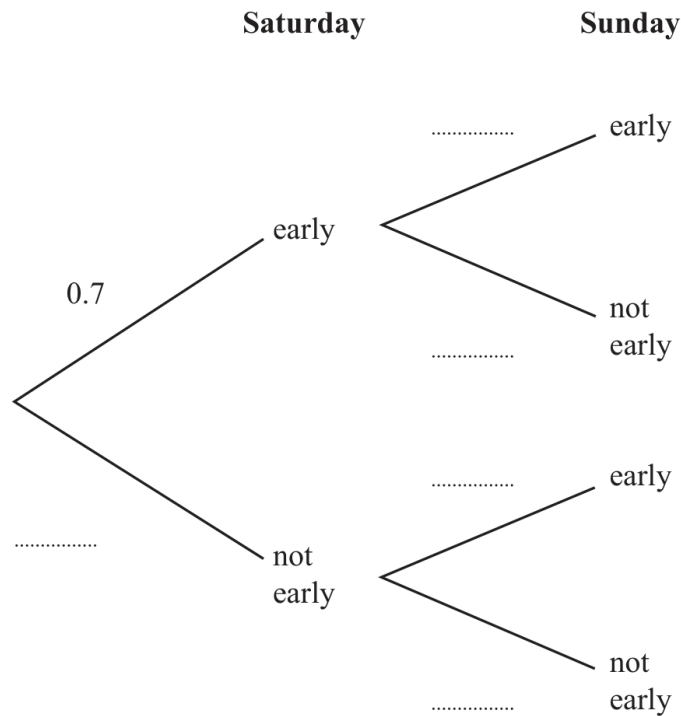
4 marks

- 13 The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

- (a) Use this information to complete the probability tree diagram.



(2)

- (b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

.....
(2)

Worked solution 10

Complete a conditional probability tree / Probability of being early both days

MODULAR UNIT 1**4 marks****M01 – Complete a conditional probability tree****Complete Saturday branches**

$$P(\text{early}) = 0.7, \quad P(\text{not early}) = 0.3.$$

Complete Sunday branches

After early: 0.9 and 0.1; after not early: 0.6 and 0.4.

Final answer

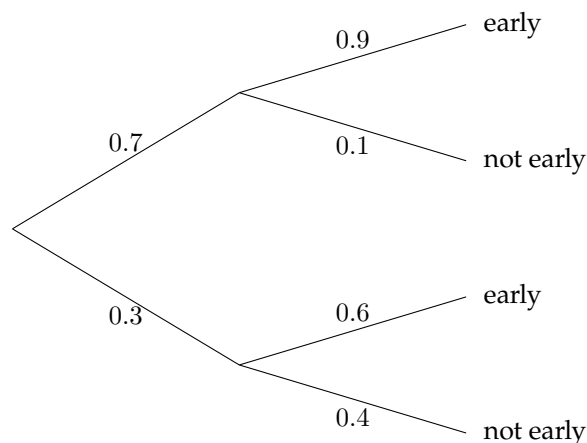
Branches: 0.7/0.3, then 0.9/0.1 and 0.6/0.4

M02 – Probability of being early both days**Multiply along the early path**

$$P(\text{early Saturday and early Sunday}) = 0.7 \times 0.9 = 0.63.$$

Final answer

0.63

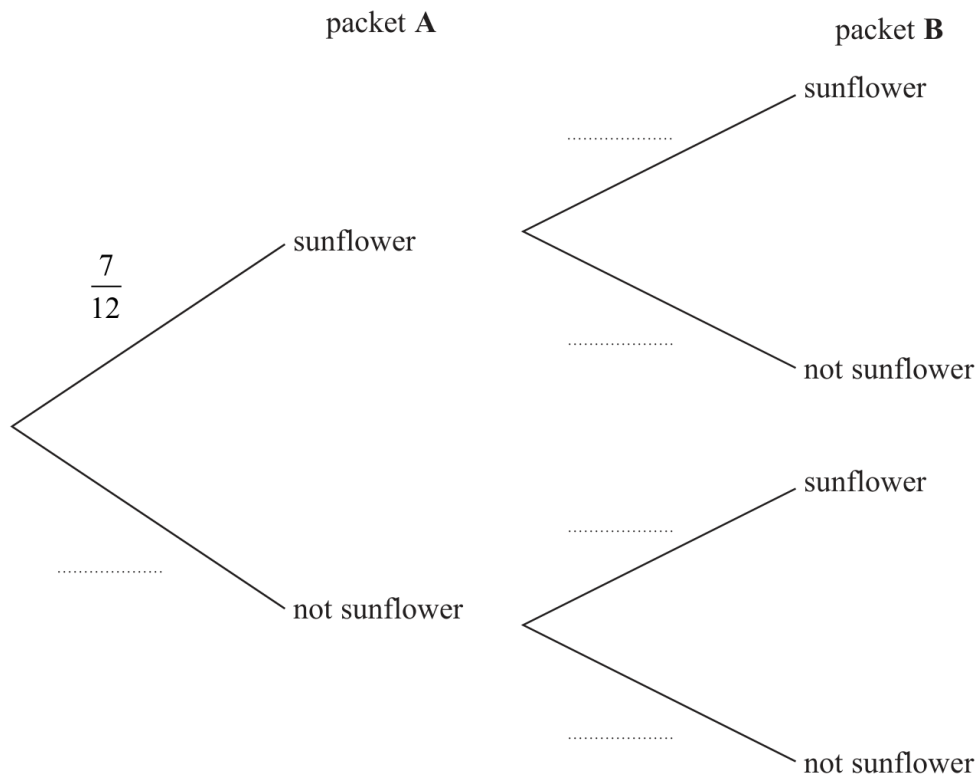
Completed solution diagram: Completed conditional probability tree

Original question 12

4MA1-1H Q14 Q14(a)

ORIGINAL QUESTION**2 marks****14** Aika has 2 packets of seeds, packet **A** and packet **B**There are 12 seeds in packet **A** and 7 of these are sunflower seeds.There are 15 seeds in packet **B** and 8 of these are sunflower seeds.Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(2)

Worked solution 12

Chapter: Probability Diagrams - Venn & Tree Diagrams

MODULAR UNIT 1

2 marks

Q14(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Construct and read probability trees • Variant: Complete an independent-event tree

Method

From packet A the branches are sunflower $7/12$ and not sunflower $5/12$; from packet B they are sunflower $8/15$ and not sunflower $7/15$.

Why: Use the given seed counts to complete all four branches.

Method

Place these four probabilities on the supplied tree.

Why: The second draw is from a different packet, so the branch probabilities remain unchanged.

Final answer

tree probabilities: $7/12$, $5/12$, $8/15$, $7/15$

Completed diagram 12

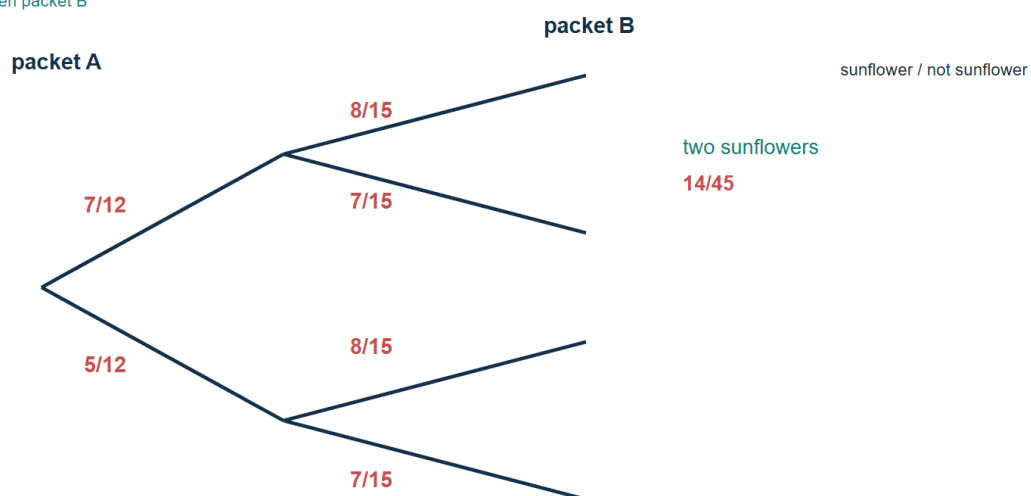
Chapter: Probability Diagrams - Venn & Tree Diagrams

MODULAR UNIT 1

2 marks

Q14 • Completed probability tree

Packet A then packet B



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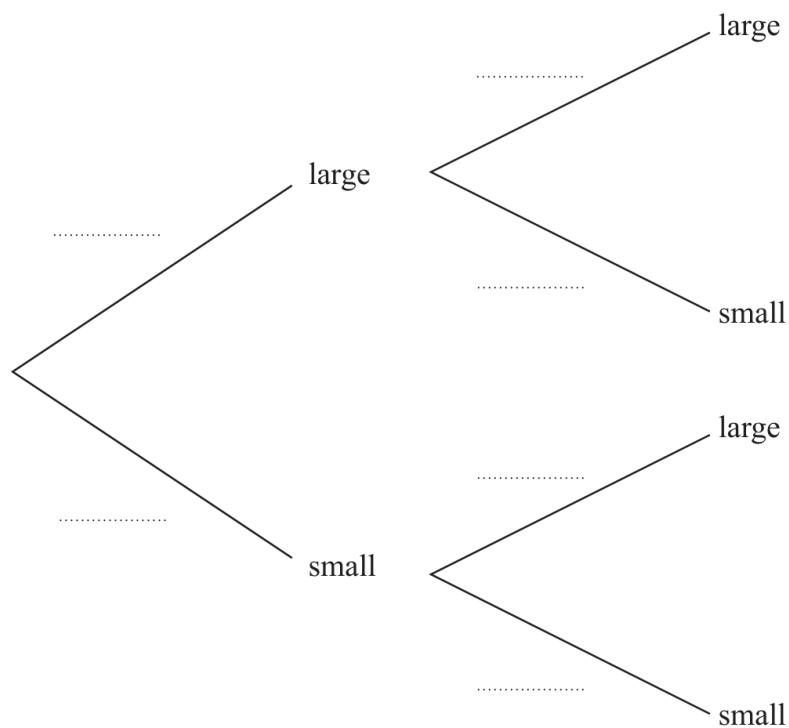
14 There are 20 glasses in a cupboard.

13 of the glasses are large

7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

(2)

Worked solution

January 2020 | Question 14 | 2 marks

Family: Construct and read probability trees • Variant: Complete a without-replacement tree

Method / working

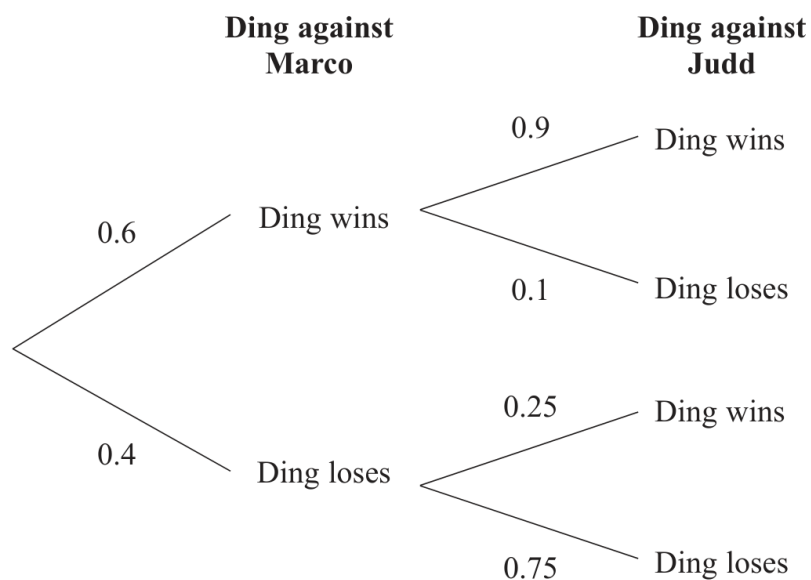
1. First draw: large= $\frac{13}{20}$, small= $\frac{7}{20}$.

2. Without replacement: after large, large= $\frac{12}{19}$ and small= $\frac{7}{19}$; after small, large= $\frac{13}{19}$ and small= $\frac{6}{19}$.

Final answer: Tree: $\frac{13}{20}$, $\frac{7}{20}$; then $\frac{12}{19}$, $\frac{7}{19}$ and $\frac{13}{19}$, $\frac{6}{19}$

- 14 Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.



- (a) Work out the probability that Ding will win both games.

(2)

Worked solution

January 2021 | Question 14 | 2 marks

Family: Construct and read probability trees • Variant: Read one complete path from a probability tree

Method / working

1. Multiply the first-game win probability by the second-game win probability: 0.6×0.9 .
2. The result is 0.54.

Final answer: 0.54

(a) Complete the probability tree diagram.



If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

Worked solution

June 2021 | Question 13 | 2 marks

Family: Construct and read probability trees • Variant: Complete an independent-event tree

Method / working

1. Draw first-race branches $W=.7$ and not $W=.3$.
2. From each branch draw second-race $W=.4$ and not $W=.6$.

Final answer: Tree: first win $.7/.3$; second win $.4/.6$

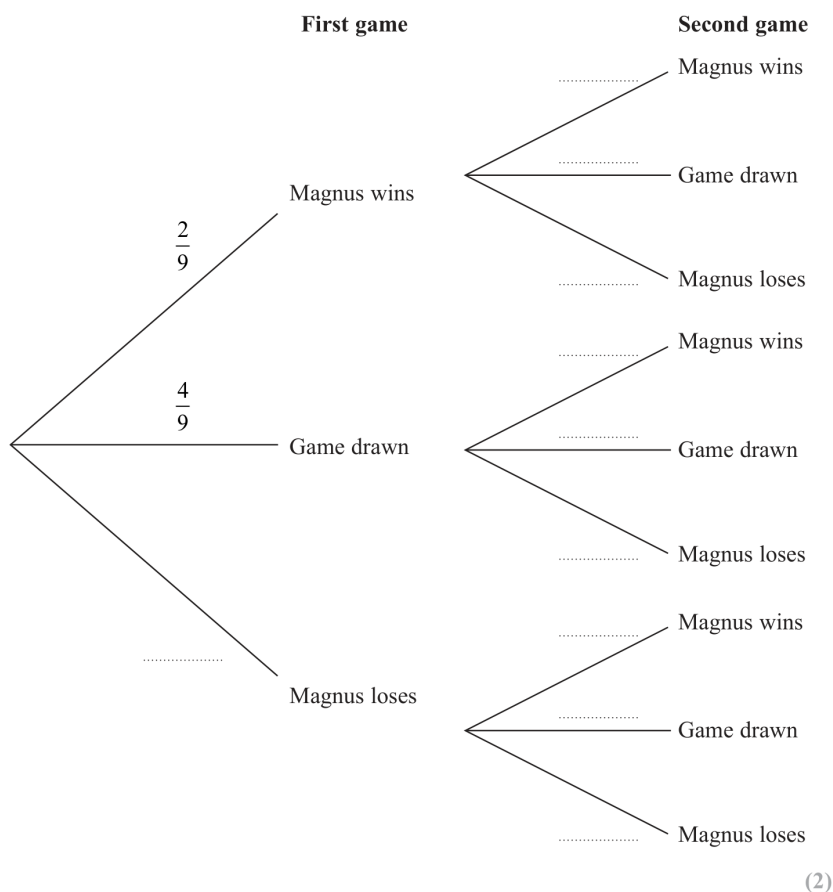
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(3)

Worked solution

November 2021 | Question 15 | 2 marks

Family: Construct and read probability trees | Variant: Complete an independent-event tree

METHOD / WORKING

1. For each chess game:

2. $P(\text{M wins}) = \frac{2}{9}$, $P(\text{draw}) = \frac{4}{9}$, $P(\text{M loses}) = \frac{1}{3}$

3. After two games, equal points happens if both games are drawn or Magnus wins one and loses one.

4. $\left(\frac{4}{9}\right)^2 + 2\left(\frac{2}{9}\right)\left(\frac{1}{3}\right)$ 5. $= \frac{16}{81} + \frac{12}{81} = \frac{28}{81}$

6. After three games, equal points means Magnus scores 3 points in total.

7. This happens with three draws, or with one win, one draw and one loss.

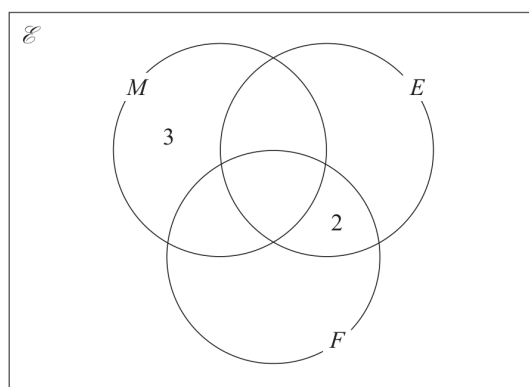
8. $\left(\frac{4}{9}\right)^3 + 6\left(\frac{2}{9}\right)\left(\frac{4}{9}\right)\left(\frac{1}{3}\right)$ 9. $= \frac{64}{729} + \frac{144}{729} = \frac{208}{729}$ 10. Answers: $\frac{28}{81}$, $\frac{208}{729}$

FINAL ANSWER

 $\frac{28}{81}$, $\frac{208}{729}$

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

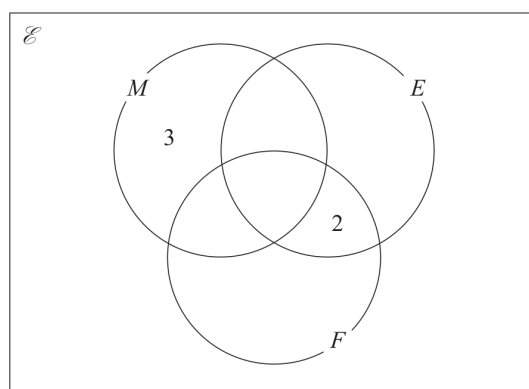
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

Worked solution

November 2021 | Question 16 | 2 marks

Family: Represent and read events with Venn diagrams | Variant: Solve unknown Venn regions using a total

METHOD / WORKING

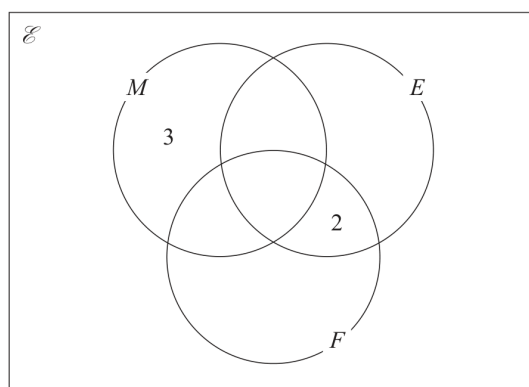
1. Use the Maths circle.
2. $M \text{ only} = 3$
3. $M \cap E \text{ only} = 18 - x$
4. $M \cap F \text{ only} = 11 - x$
5. The total in M is 25:
6. $3 + (18 - x) + (11 - x) + x = 25$
7. $32 - x = 25$
8. $x = 7$
9. So:
10. $M \cap E \text{ only} = 11$, $M \cap F \text{ only} = 4$, $E \cap F \text{ only} = 2$
11. $E \text{ only} = 20 - 11 - 2 = 7$
12. $F \text{ only} = 14 - 4 - 2 = 8$
13. For a student who passed Maths, failed French means M but not F :
14. $\frac{3 + 11}{25} = \frac{14}{25}$
15. Answers: $x = 7$, and $\frac{14}{25}$.

FINAL ANSWER

 $x = 7$, and $\frac{14}{25}$.

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
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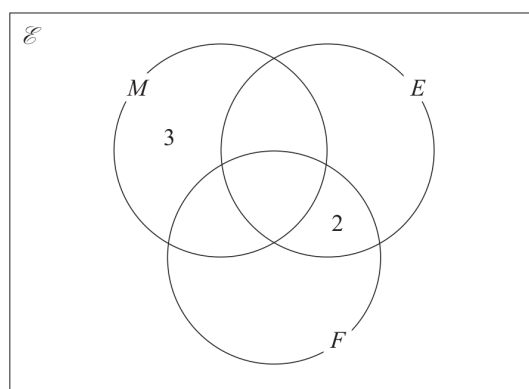
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- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

Worked solution

November 2021 | Question 16 | 2 marks

Family: Represent and read events with Venn diagrams | Variant: Complete a three-set Venn diagram

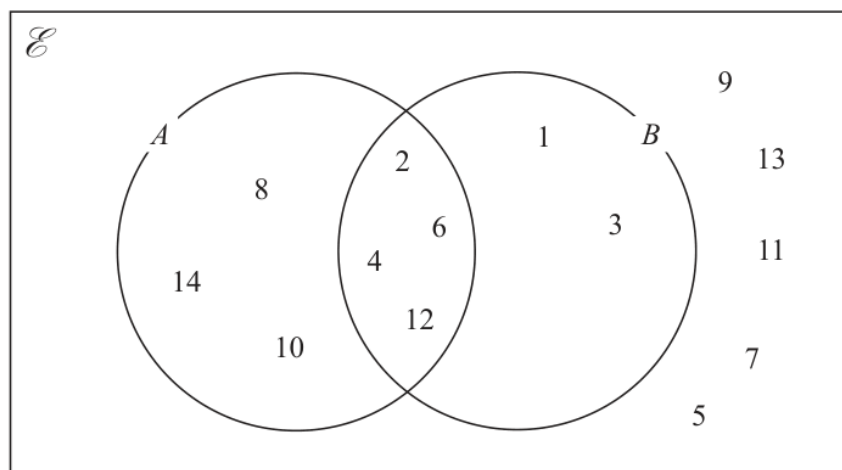
METHOD / WORKING

1. Use the Maths circle.
2. $M \text{ only} = 3$
3. $M \cap E \text{ only} = 18 - x$
4. $M \cap F \text{ only} = 11 - x$
5. The total in M is 25:
6. $3 + (18 - x) + (11 - x) + x = 25$
7. $32 - x = 25$
8. $x = 7$
9. So:
10. $M \cap E \text{ only} = 11$, $M \cap F \text{ only} = 4$, $E \cap F \text{ only} = 2$
11. $E \text{ only} = 20 - 11 - 2 = 7$
12. $F \text{ only} = 14 - 4 - 2 = 8$
13. For a student who passed Maths, failed French means M but not F :
14. $\frac{3 + 11}{25} = \frac{14}{25}$
15. Answers: $x = 7$, and $\frac{14}{25}$.

FINAL ANSWER

 $x = 7$, and $\frac{14}{25}$.

- 1 The numbers from 1 to 14 are shown in the Venn diagram.



- (a) List the members of the set $A \cap B$

.....
(1)

- (b) List the members of the set B'

.....
(1)

A number is picked at random from the numbers in the Venn diagram.

- (c) Find the probability that this number is in set A but is **not** in set B.

.....
(2)

(Total for Question 1 is 4 marks)

Answer reference | May-Nov 2020 | Question 1

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Q	WORKING	ANSWER	MARK	NOTES
1 (a)		2, 4, 6, 12	1	B1
(b)		5, 7, 8, 9, 10, 11, 13, 14	1	B1
(c)			2	M1 for $\frac{a}{14}$ with $a < 14$ or $\frac{3}{b}$ with $b > 3$ or for 3 and 14 used with incorrect notation e.g. 3 : 14
		$\frac{3}{14}$		A1 for $\frac{3}{14}$ or 0.214(...)
Total 4 marks				

Worked solution

May-Nov 2020 | Question 1 | 4 marks

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Solution

From the Venn diagram,

$A \cap B = \{2, 4, 6, 12\}$

The values not in B are

$B' = \{5, 7, 8, 9, 10, 11, 13, 14\}$

The values in A but not in B are

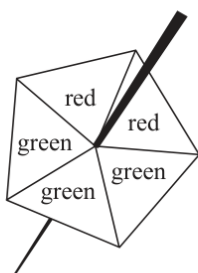
$\{8, 10, 14\}$

So

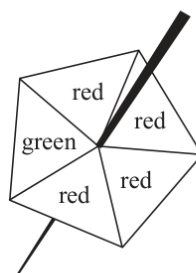
$P(A \cap B) = \frac{3}{14}$

Answers: $\{2, 4, 6, 12\}$, $\{5, 7, 8, 9, 10, 11, 13, 14\}$, $\frac{3}{14}$.

15 Harry has two fair 5-sided spinners.



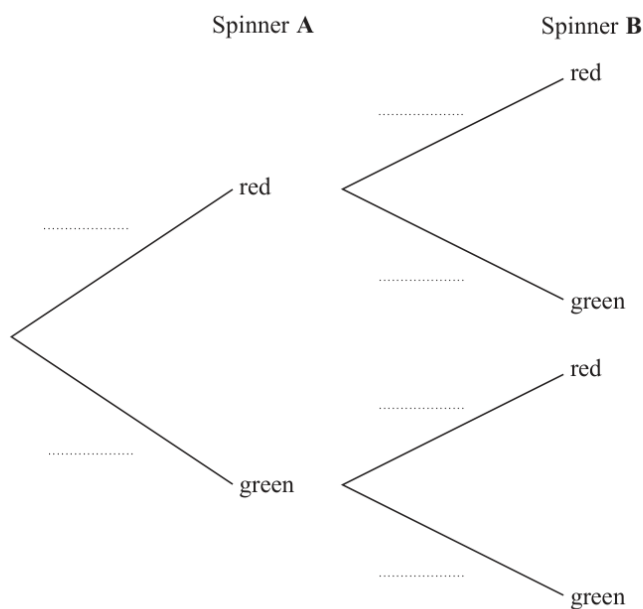
Spinner A



Spinner B

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(2)



(b) Work out the probability that at least one of the spinners will land on green.

.....
(3)

(Total for Question 15 is 5 marks)

Answer reference | May-Nov 2020 | Question 15

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

15	(a)		$\frac{2}{5}, \frac{3}{5}$ oe	2	B1 correct probabilities for spinner A
			$\frac{4}{5}, \frac{1}{5}, \frac{4}{5}, \frac{1}{5}$ oe		B1 correct probabilities for spinner B
	(b)	$\frac{2}{5} \times \frac{4}{5} \left(= \frac{8}{25} \right)$ or $\frac{2}{5} \times \frac{1}{5} \left(= \frac{2}{25} \right)$ or $\frac{3}{5} \times \frac{4}{5} \left(= \frac{12}{25} \right)$ or $\frac{3}{5} \times \frac{1}{5} \left(= \frac{3}{25} \right)$ oe		3	M1 ft from (a) provided $0 < \text{probability} < 1$
		$1 - \frac{8}{25}$ or $\frac{2}{25} + \frac{12}{25} + \frac{3}{25}$ or $\frac{2}{25} + \frac{3}{5}$ oe			M1 ft from (a) for a complete method
			$\frac{17}{25}$		A1 oe
					Total 5 marks

Worked solution

May-Nov 2020 | Question 15 | 5 marks

Family: Represent and read events with Venn diagrams | Variant: Complete a two-set Venn diagram from totals

METHOD / WORKING

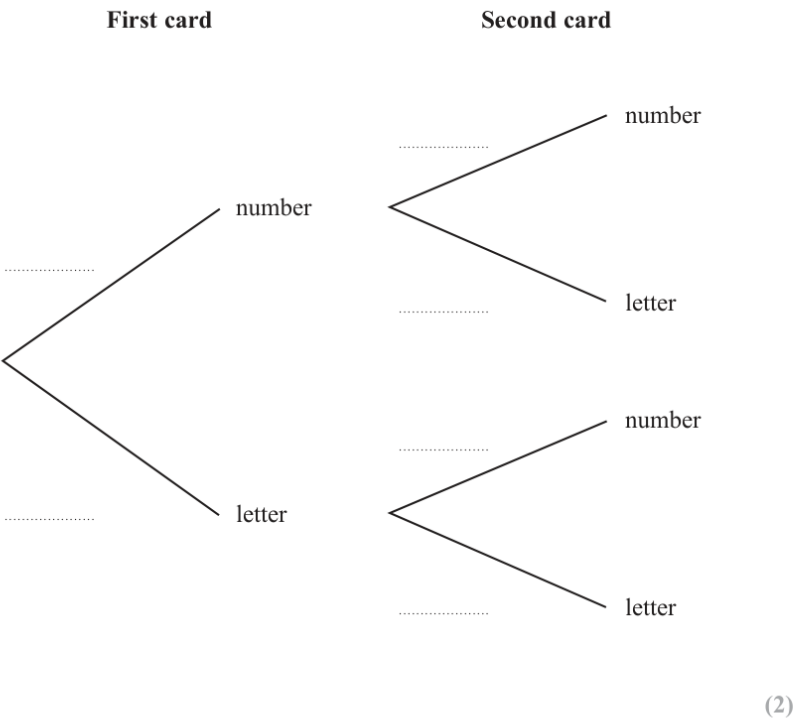
Worked solution
May-Nov 2020 Question 15 5 marks
Family: Represent and read events with Venn diagrams Variant: Complete a two-set Venn diagram from totals
Solution
Spinner A has 2 red sectors and 3 green sectors:
$P(A_R) = \frac{2}{5}, \text{ and } P(A_G) = \frac{3}{5}$
Spinner B has 4 red sectors and 1 green sector:
$P(B_R) = \frac{4}{5}, \text{ and } P(B_G) = \frac{1}{5}$
At least one green is the complement of both red.
$P(\text{at least one green}) = 1 - \frac{2}{5} \times \frac{4}{5}$
$= 1 - \frac{8}{25} = \frac{17}{25}$
Answer: $\frac{17}{25}$.

FINAL ANSWER

15 Here are 9 cards. Each card has either a number on it or a letter on it.



Tomas is playing a game.
 Tomas takes at random one of the cards and keeps it.
 Tomas then takes at random another card and keeps it.
 (a) Complete the probability tree diagram.



(b) Work out the probability that each of the two cards has a number on it.

.....
(2)

(c) Work out the probability that there will be one card with a number on it and one card with a letter on it.

.....
(3)

.....
(Total for Question 15 is 7 marks)

Answer reference | January 2023 | Question 15

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

15	(a)		$\frac{2}{9}, \frac{7}{9}$	2	B1 for correct probabilities for the first card Allow equivalent probabilities e.g. $0.\dot{2}$
			$\frac{1}{8}, \frac{7}{8}, \frac{2}{8}, \frac{6}{8}$		B1 for correct probabilities for the second card Allow equivalent probabilities
	(b)	$\frac{2}{9} \times \frac{1}{8}$ or $1 - \frac{2}{9} \times \frac{7}{8} - \frac{7}{9} \times \frac{2}{8} - \frac{7}{9} \times \frac{6}{8}$		2	M1ft (All probabilities must be less than 1)
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{1}{36}$		A1ft oe probability must be less than 1 Allow equivalent decimal to at least 2 sf (truncated or rounded) for $\frac{1}{36}$ (= 0.027(77..))
	(c)	$\frac{2}{9} \times \frac{7}{8}$ or $\frac{7}{9} \times \frac{2}{8}$ oe or $\frac{2}{9} \times \frac{1}{8}$ and $\frac{7}{9} \times \frac{6}{8}$ oe or $\frac{1}{36}$ and $\frac{7}{9} \times \frac{6}{8}$ oe		3	M1ft (All probabilities must be less than 1)
		$\frac{2}{9} \times \frac{7}{8} + \frac{7}{9} \times \frac{2}{8}$ or $2 \times \frac{14}{72}$ oe or $1 - \frac{2}{9} \times \frac{1}{8} - \frac{7}{9} \times \frac{6}{8}$ oe or $1 - \frac{1}{36} - \frac{7}{9} \times \frac{6}{8}$ oe			M1ft

Answer reference | January 2023 | Question 15

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{7}{18}$	A1ft oe probability must be less than 1 Allow equivalent decimal to at least 2 sf (truncated or rounded) for $\frac{7}{18}$ (= 0.38(88...))
			Total 7 marks

Worked solution

January 2023 | Question 15 | 7 marks

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

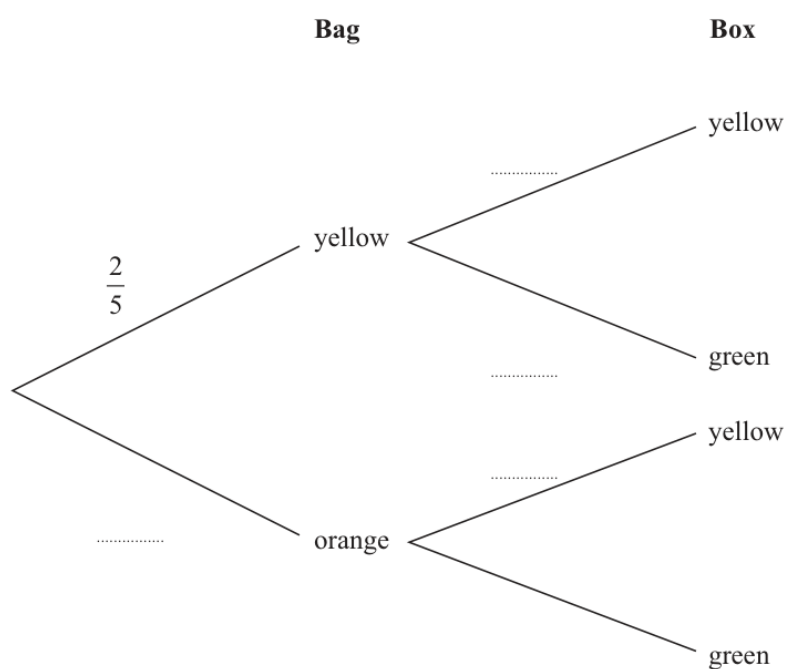
Official final mark-scheme pages are embedded immediately after this question.

13 In a bag, there are only 2 yellow beads and 3 orange beads.

In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Chi takes two beads of different colours.

(3)

(Total for Question 13 is 5 marks)

Worked solution

November 2025 | Question 13 | 5 marks

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

Solution

Bag 1:

$$P(Y) = \frac{25}{100}, \text{ and } P(O) = \frac{35}{100}$$

Bag 2:

$$P(Y) = \frac{38}{100}, \text{ and } P(G) = \frac{58}{100}$$

The only way to get the same colour is yellow then yellow.

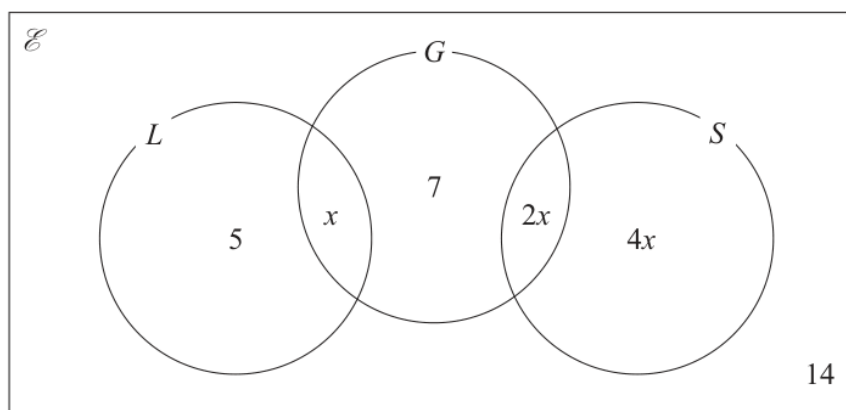
$$P(\text{different}) = 1 - \frac{25}{100} \times \frac{38}{100}$$

$$= 1 - \frac{3}{20} = \frac{17}{20}$$

Answer: $\frac{17}{20}$.

- 18 Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.

Work out the probability that this student studies Latin or Sanskrit.

(Total for Question 18 is 3 marks)

Worked solution

November 2025 | Question 18 | 3 marks

Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals

Solution

The total is 40.

The regions shown add to

$$5+x+7+2x+4x+14$$

So

$$26+7x=40$$

$$7x=14$$

$$x=2$$

Latin or Sanskrit:

$$5+x+2x+4x=5+2+4+8=19$$

$$P(\text{L union S})=(19)/(40)$$

Answer: $(19)/(40)$.